

CEE 406 Pavement Design I
Department of Civil and Environmental Engineering
University of Illinois at Urbana-Champaign
Homework 8
Due: 11/12/2024

Problem 1

Design a new rural 4-lane State highway in central Sangamon County in Illinois. The desired performance period is 20 years.

- ADT (Design Year Traffic)
 - PV = 15,000
 - SU = 320
 - MU = 300
- Subgrade Support Rating (SSR) = Fair
- Asphalt Binder Type = PG64-22

Perform a flexible mechanistic design for new construction.

Problem 2

Consider the pavement system provided in Figure 1. AC layer is divided into 5 sublayers. The AC moduli used for the sublayers are 570,000psi, 565,000psi, 575,000psi, 595,000psi, 585,000psi **in no order**.

- a) Assume a 6000lb single wheel load is applied at a 6-in radius are. Please calculate the **tensile strains** and **compressive strains** in the **middle of each layer/sublayer** and at **12in below subgrade interface**, under the center of the load. Using WinJULEA, please organize the AC moduli for each layer based on the frequency each layer is exposed to (assume temperature is constant through the depth). Please provide screenshots of the input and report results.
- b) If the 6000lb load is applied 1000 times a day for 90 days, using AASHTOWare procedure, please calculate the AC **bottom-up cracking** and **rutting** and **base and subgrade rutting**. For subgrade, please use a thickness of 24in to calculate rutting. Please do NOT subdivide the base layer and subgrade. Please plot total rutting and cracking with respect to number of repetitions and describe the progression of these distresses.
- c) What is the total rutting of the system? Which layer contributes to rutting the most?

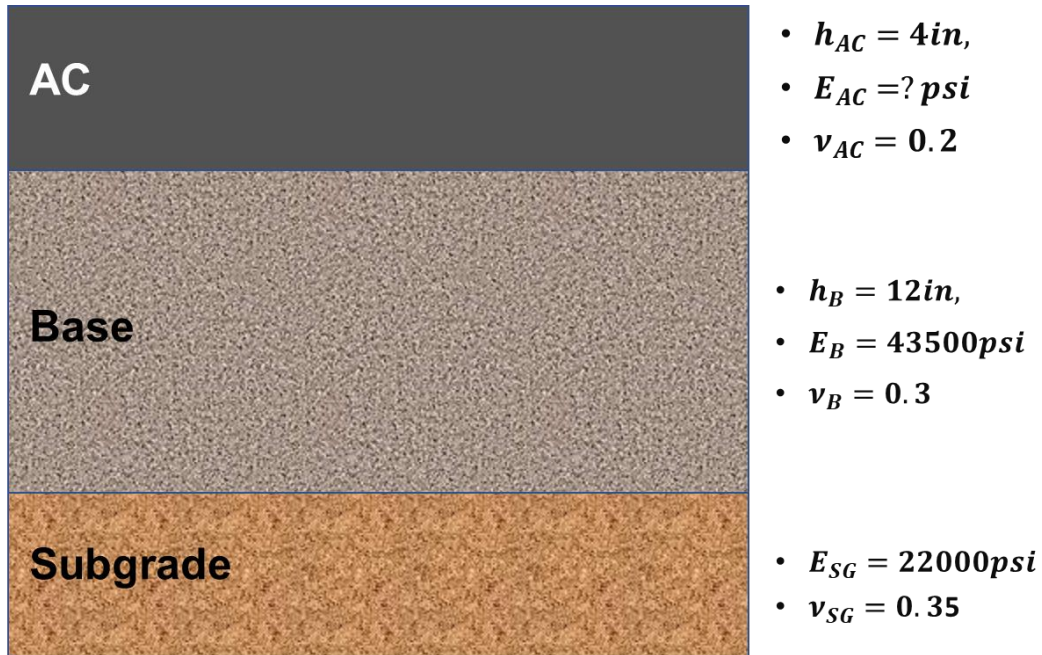


Figure 1 Pavement System

Table 1 AC sublayer information

AC Sublayer Thickness (in)	AC Sublayer Depth (in)	Modulus (psi)
0.5	0.25	
0.5	0.75	
1	1.5	
1	2.5	
1	3.5	

Rutting Equations for AC

$$Rut = \epsilon_v * h_{AC} * 3.5 * 10^{-3.4488} * T^{1.5606} * N^{0.479244}, \quad (in)$$

ϵ_v = strain in the middle of sublayer,

h_{AC} = thickness of AC in inches,

T = AC Temp in °F, assume 71°F,

N = number of repetitions

Rutting Equations for Base and Subgrade

$$Rut = \epsilon_v * h * 46.55 * \exp\left(-\left(\frac{10785.6}{N}\right)^{0.174}\right), \quad (in)$$

ϵ_v = strain in the middle of sublayer,

h = thickness in psi,

N = number of repetitions

Cracking Equations

$$FC_{bottom} = \frac{1}{60} * \frac{6000}{1 + \exp(1.6193 - 2.81 * \log_{10}(DI_{bottom} * 100))}$$

FC_{bottom} = bottom-up cracking in %,

$$DI = \sum \frac{n}{N_f}$$

n = number of repetitions

$$N_f = 0.003612 * C_H * \epsilon_t^{-3.9492} * E_{AC}^{-1.281}$$

N_f = number of allowable repetitions,

ϵ_t = tensile strain at the bottom of AC,

E_{AC} = AC modulus in psi (use the lowest sublayer modulus)

$$C_H = \frac{1}{0.000398 + \frac{0.003602}{1 + \exp(11.02 - 3.49h_{HMA})}}$$

h_{HMA} = total AC thickness in inches